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Dr Ellie Murray, ScD

@EpiEllie

Let's start some research methods controversy with my second rule of asking good causal questions!

If you want to ask good (quantitative) research questions about cause and effect, then Rule 2 of asking good questions is:

Choose modifiable exposures.

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...

If you're not a causal inference person, that might seem fairly reasonable. But this is actually a hugely controversial issue!

Because "modifiable" implies different things to different people.

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Dr Ellie Murray, ScD @EpiEllie · Nov 18

...

So what do "I" mean by "modifiable"?

I mean that the cause you are interested in must sometimes change in at least some situations.

You cannot ask a quantitative causal question about a cause which never changes. You will never have data!

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...

But, I don't mean that you yourself have to be able to modify the cause!

If the cause is something humans can directly modify, then "interpreting" your quantitative causal estimate is easier.

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But "estimating" a quantitative causal effect can be done even when we can't modify the cause ourselves.

It's just that the "interpretation" becomes more difficult. And the specific type of causal effect you are estimating changes.

(for details see ncbi.nlm.nih.gov/pmc/articles/P...)

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One last thing to know about Rule 2:

If we are interested in "qualitative" questions about cause and effect, then we do not necessarily need the cause to ever have changed.

But we "do" need to be able to "imagine" what change in that cause might look like!

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Dr Ellie Murray, ScD @EpiEllie · Nov 18

Want to learn more about asking good causal questions?

Last week I shared Rule 1 🙋

That rule? Be SPECIFIC!



Dr Ellie Murray, ScD @EpiEllie · Nov 11

How do we do good research on causes and effects?

Rule number 1: be specific about the questions you are asking!

What's a specific causal question? One that can help inform a specifi...

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